**INDIAN INSTITUTE OF TECHNOLOGY BOMBAY**

*Department of Mathematics*

*SI 427 (Probability I)*

**Tutorial Sheet-VIII**

1. Let $X$ be a non negative integer valued random variable such that mean of $X$ is finite. Show that
$$EX \;=\; \sum_{n=1}^{\infty} P\big(X \geq n\big)\,.$$

2. Let $X$ be a discrete random variable with pmf $f$ and $\varphi : \mathbb{R} \to \mathbb{R}$ be a function. Show that
$$E[\varphi \circ X] \;=\; \sum_{x \in D} \varphi(x) f(x),$$
provided the rhs series converges absolutely, where $D \;=\; \big\{x \in \mathbb{R} | P\big(X = x\big) > 0\big\}$.

3. Let $X_1, X_2, \cdots, X_n$ be iid Bernoulli($\theta$) and $\hat{\theta}_n \;=\; \frac{1}{n}\sum_{i=1}^{n} X_i$. Show that $\theta_n$ is an unbiased estimator for $\theta$.[1]

4. Give an example of discrete random variable infinite mean.

5. Let $X$ be a Binomial random variable with parameters $n = 4,\ p = \frac{1}{2}$. Find $E \sin \frac{\pi X}{2}$.

6. Let $X$ be Poisson with parameter $\lambda$. Find the mean of $\frac{1}{1+X}$.

7. Let $X$ be a geometrically distributed [2] random variable with parameter $p$ and $M$ be a positive integer. Find $E \min\{X, M\}$.

8. Let $X$ be a nonnegative continuous random variable with finite mean. Show that
$$EX \;=\; \int_0^{\infty} P\big(X > t\big)\, dt\,.$$

9. Let $\varphi(x) = \lfloor x \rfloor$ and $X$ is exponential (1). Find the pmf of $\varphi(X)$ and hence find $E[\varphi(X)]$.

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[1] i.e. $E\theta_n = \theta$

[2] A random variable $X$ is said be geometric with parameter $0 < p < 1$ if its pmf is given by $f(x) \;=\; p(1-p)^x,\ x = 0, 1, \ldots$